Applications of Calculus

Questions 1–8

Question 1.

Applications of Differentiation

For each of the following situations, find the first differential, explain what it means in context and specify its units:

a) $C = 5.8x + 14.2$ where £C is the cost of the transport and x is the miles travelled.
b) $h = -t^2 + 6t + 135$ at $t = 10$ where h is the height of a ball (in metres) and t is the number of seconds passed.
c) $T = 0.01t^3 - 0.2t^2 + 20$ at $t = 5$ where T is the temperature (°C) of a substance and t is the number hours that have passed.

Question 2. The spread of a virus over a period of 22 days may be modelled by the equation

$$N = -0.49t^2 + 10t + 3.54$$

where N is the number of people with the virus and t is the number of days passed. Find

a) the rate of spread of the virus on day 5
b) the rate of spread of the virus on day 15
c) the coordinate of the maximum of the curve of the model giving the points to 3 significant figures.
d) explain the meaning of the maximum point you found in part c)

Question 3. The height h of a seedling in mm is given by

$$h = -0.028n^2 + 1.95n + 6.2$$

where n is the number of days passed.

a) Find $\frac{dh}{dn}$ and state its physical relevance.
b) Use this model to predict the day at which the seedling reaches maximum height and state its height.
c) How effective is the model in the long term?

Question 4. The velocity of an object over t seconds may be given by

$$v = \frac{7}{3}t^3 - 8t^2 + 4t + 10, \qquad 0 \leq t \leq 3$$

where v is velocity in metres per second.

a) Find the coordinates of the stationary points of the model giving your answer to 3 significant figures.
b) Explain the meaning of stationary points in the context of the object.

Question 5. The depth of water y cm below a model boat at time t seconds is modelled by

$$y = t^3 - 8t^2 + 15t + 48, \qquad 0 \leq t \leq 5$$

a) Find the greatest depth of water below the model boat.
b) Find the minimum value of $\frac{dy}{dt}$, the time at which it occurs and the depth of the water at this time.
c) State the vertical direction the boat is moving at the time found in part b)

Question 6. A model for the height h of a roller coaster in feet is

$$h(x) = 2x^3 - 30x^2 + 94x + 150$$

where x is the horizontal distance in hundreds of feet. Find
a) the gradient of $h(x)$ when $x = 0$
b) the angle of inclination (in degrees) to the ground of the rollercoaster initially.
c) the highest point of the rollercoaster and where they occur, giving your answer to the nearest feet.

Question 7.

Applications of Integration

The height h of a pile of sand above ground level can be modelled by

$$h(x) = -10x^3 + 14x^2 + \frac{3}{2}x$$

where x is the number of metres from one side. The mean height of the pile of sand may be given by

$$A = \frac{1}{2}\int_0^2 h(x)\,dx$$

Find the value of A

Question 8. The value of some shares £V during the first 12 months of the year may be modelled by

$$V(t) = 0.5t^3 + 3.5t^2 - 12t + 50$$

where t is the number of months passed.

a) Use this model to predict the value of the shares when 12 months have passed.

The mean share of the shares value may be given by

$$A = \frac{1}{12}\int_0^{12} V(t)\,dt$$

b) Find the value of A